

C01 AT3 LEVEL 1



Secure Exit

SIMATS Online Programming Lab
C++ PROGRAMMING

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QUESTIONS

LEVEL 1 — EASY

Q1

Number

✓

Q2

Number

✓

Q3

Number

✓

Q4

Pattern

✓

4 / 4 saved

Level 1 — Easy hexadecimal (E) — Number

✓ Submitted

Develop a program that converts a decimal number provided by the user into its hexadecimal equivalent

Example Input:

Enter a decimal number: 255

Example Output:

Hexadecimal equivalent: FF

Test Cases:

1.55
2.100
3.151
4.2022

Your Code

```
1 #include<iostream>
2 #include<iomanip>
3 using namespace std;
4 int main()
5 {
6     int n;
7     cin>>n;
8     cout<<uppercase<<hex<<n;
9     return 0;
10 }
```

Output

```
FF
```



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LEVEL 1 — EASY

Q1

Number

✓

Q2

Number

✓

Q3

Number

✓

Q4

Pattern

✓

4 / 4 saved

Level 1 — Easy sum (E) — Number

✓ Submitted

Develop a program that calculates the sum of positive and negative numbers separately from a list of numbers provided by the user.

Example Input:

Enter number of elements: 5
Enter elements: -3 4 -2 7 -5

Example Output:

Sum of positive numbers: 11
Sum of negative numbers: -10

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int n,num;
6     int positivesum=0,negativesum=0;
7     cin>>n;
8     for(int i=1;i<=n;i++){
9         cin>>num;
10        if(num>0)
11            positivesum+=num;
12        else if (num<0)
13            negativesum+=num;
14    }
15    cout<<"sum of positive numbers:"<<positivesum<<endl;
```

Output

```
sum of positive numbers:11
sum of negative numbers:-10
```



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QUESTIONS

LEVEL 1 — EASY

Q1

Number

✓

Q2

Number

✓

Q3

Number

✓

Q4

Pattern

✓

4 / 4 saved

Level 1 — Easy

automorphic (E) — Number

✓ Submitted

Create a C program to determine if a number is an automorphic number. (An automorphic number is a number whose square ends in the same digits as the original number.)
Example Input:
Enter a number: 25

Example Output:

25 is an automorphic number

Test Cases:
1.5
2.10
3.15
4.20

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int n,temp,square,digits=1;
6     cin>>n;
7     square=n*n;
8     temp=n;
9     while(temp>=10){
10         digits*=10;
11         temp/=10;
12     }
13     digits*=10;
14     if(square%digits==n)
```

Output

```
25 is an automorphic number
```



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QUESTIONS

LEVEL 1 — EASY

Q1

Number

✓

Q2

Number

✓

Q3

Number

✓

Q4

Pattern

✓

4 / 4 saved

Level 1 — Easy

Spiral Pattern (E) — Pattern

✓ Submitted

Create a program to Print the following Spiral Pattern

```
1 2 3 4 5
16 17 18 19 6
15 24 25 20 7
14 23 22 21 8
13 12 11 10 9
```

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     int a[5][5]={
6         {1,2,3,4,5},
7         {16,17,18,19,6},
8         {15,24,25,20,7},
9         {14,23,22,21,8},
10        {13,12,11,10,9}
11    };
12    for(int i=0;i<5;i++){
13        for(int j=0;j<5;j++){
14            cout<<a[i][j]<<" ";
15        }
16        cout<<endl;
17    }
18    return 0;
19 }
```

Output

```
1 2 3 4 5
16 17 18 19 6
15 24 25 20 7
14 23 22 21 8
13 12 11 10 9
```